

PROBLEM STATEMENT #33 | Retail, E-Commerce & Finance**Micro-Lending & Peer Credit Risk Assessor****1. Problem Context & Overview**

A peer-to-peer micro-finance platform calculating borrower risk scores using heuristic rules, generating structured repayment schedules, and triggering automated default warnings.

Target Stakeholders / Actors: *Borrower, Peer Lender*

2. Sample Functional Requirement (FR) Guideline**Sample Requirement: FR-001 [Priority: High]**

Description: The system shall compute borrower credit risk ratings based on debt-to-income ratio, repayment history, and employment stability indicators.

Sample Acceptance Criteria: Pass: Risk tier (Low/Med/High) accurately assigned; Fail: Incomplete financial profile generates credit approval.

3. Sample Non-Functional Requirement (NFR) Guideline**Sample Requirement: NFR-001 [Type: Performance & Security]**

Description: Loan repayment transactions and EMI distribution must execute inside database ACID transactions to prevent double-spending.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]**• Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

• UML Use-Case Diagram:

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

• Use-Case Flow Specification:

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.